

BREAST TUMOR MOLECULAR PROFILE — HOTSPOT PANEL + ERBB2 CN — FINAL REPORT

PATIENT

BERGSTROM, ANITA L.
MRN MRN-5519082
52 y / Female

SPECIMEN

Lab accession WCI-26-MOL-08812
Client accession WCI-26-S-04115
FFPE block A4 (tumor)
Collected 2026-04-22

ORDERING

Westhaven Cancer Institute
T. Okafor, MD
Received 2026-04-25
Reported 2026-05-02

CLINICAL HISTORY (PROVIDED BY ORDERING SITE)

52-year-old postmenopausal woman, gravida 2 para 2. Screening mammography (annual) demonstrated a new 2.1 cm spiculated mass in the upper outer quadrant of the left breast (BI-RADS 5). Core needle biopsy showed invasive carcinoma. No personal history of breast or ovarian cancer. Mother diagnosed with breast cancer at age 68 (receptor status unknown). Maternal aunt with ovarian cancer at 62. Patient elected breast-conserving surgery with sentinel lymph node biopsy on 2026-04-22.

RESULT SUMMARY

POSITIVE — **PIK3CA p.(His1047Arg)** hotspot variant detected (Tier I, ACTIONABLE for PI3K-targeted therapy in advanced/metastatic setting per current NCCN). **ERBB2 copy number 2.1** (diploid, no amplification; concordant with HER2 IHC 1+ on surgical pathology). MSI: stable. TMB: 3.6 mut/Mb. See table and interpretation.

DETECTED VARIANTS

Gene	Variant (HGVS)	Consequence	VAF	Tier
PIK3CA	NM_006218.4:c.3140A>G p.(His1047Arg)	Missense, H1047 hotspot; oncogenic activation	32%	I

Tier I = strong clinical / actionable significance. PIK3CA H1047R is an FDA-recognized companion biomarker for alpelisib in combination with fulvestrant in HR-positive HER2-negative advanced/metastatic breast carcinoma after progression on endocrine therapy.

COPY-NUMBER (HER2 / ERBB2)

Gene	Copy number	Interpretation
ERBB2 (HER2)	2.1 copies	Diploid — NO amplification. Concordant with HER2 IHC 1+ (negative) on surgical pathology.

ADDITIONAL BIOMARKERS

Biomarker	Result	Cutoff / interpretation
Microsatellite Instability (MSI)	MSS (stable)	Microsatellite stable; MMR proficient (predicted).
Tumor Mutational Burden	3.6 mut/Mb	Below high-TMB threshold (10 mut/Mb).

GENES WITHOUT REPORTABLE VARIANT

No reportable variant detected in: TP53, ESR1, AKT1, PTEN, MAP2K1, BRAF (V600), KRAS, NRAS, HRAS, RB1, NF1, MAP3K1, GATA3, MED12, ARID1A, CDH1, FGFR1/2/3, MTOR, CHEK2, ATM (somatic). (Hotspot panel; for germline status of BRCA1, BRCA2, PALB2 and extended hereditary genes, see separate germline panel: ATLAS-26-G-06224.)

INTERPRETATION

The tumor harbors a **PIK3CA p.(His1047Arg) oncogenic hotspot missense variant** (VAF 32%). This is the most common activating mutation in invasive breast carcinoma and is the companion biomarker for alpelisib + endocrine combinations in advanced or metastatic HR-positive HER2-negative disease per current NCCN. **ERBB2 copy number is 2.1** — diploid — concordant with the HER2 IHC 1+ (negative) result on the surgical pathology report; no FISH is indicated. MSI is stable and TMB is below the high threshold; immune-checkpoint indications are unlikely. The PIK3CA variant is a potentially predictive (treatment-selection) biomarker. It is not a classifying feature of the WHO 5e breast carcinoma entity name and should be reported in the molecular and predictive-biomarker sections of the integrated report rather than in the diagnostic line.

METHODOLOGY & LIMITATIONS

Hotspot amplicon-based NGS panel (32 genes) on tumor DNA from FFPE block A4; mean coverage 1,400x. ERBB2 copy number assessed by panel-based read-depth ratio with orthogonal qPCR confirmation. MSI by 5-marker panel (BAT-25, BAT-26, NR-21, NR-24, MONO-27); TMB estimated from coding-region variant burden. Analytical sensitivity ~5% VAF; structural variants and large rearrangements not detected. Laboratory-developed test validated by Westhaven Cancer Institute — Molecular Diagnostics; not cleared or approved by the FDA.